No.	Solution
1(a)(i)	Using $v^2 = u^2 + 2as$, taking upward as positive, assuming no air resistance, $0 = 160^2 + 2(-9.81)s_y$ $s_y = 1305$ m Since the object reaches a height of 1100 m, lower than the calculated value of 1305 m, implying average deceleration has a magnitude that is larger than 9.81 m s^{-2}, indicating the presence of air resistance that is not negligible.
1(a)(ii)	Energy loss $= \frac{1}{2}mv^2 - mgh$ $= (0.92 \times 10^{-3})\left[\frac{160^2}{2} - (9.81)(1100)\right]$ $= 1.8483$ J $=1.85$ J
1(b)	By Newton's 2nd law, $F_{net} = ma$ $F_{air} + mg = ma$ $a = \frac{F_{air} + mg}{m}$ $= \frac{3.4 \times 10^{-3} + (0.92 \times 10^{-3})(9.81)}{0.92 \times 10^{-3}}$ $= 13.5$ m s^{-2}
1(c)	$F_A = kv^2$ where k is proportional constant.

F_A / 10^{-3} N	v / m s^{-1}	k / 10^{-7} N m^{-2} s^2
1.00	87	1.32
1.35	100	1.35
1.60	110	1.32
1.90	120	1.32
2.25	130	1.33
2.60	140	1.33
3.00	150	1.33
3.40	160	1.33

The k values for v = 87 m s^{-1}, 120 m s^{-1}, and 160 m s^{-1} are 1.32×10^{-7} N m^{-2} s^2, 1.32×10^{-7} N m^{-2} s^2, and 1.33×10^{-7} N m^{-2} s^2 respectively which are constant.

Hence shown that for speeds greater than 80 m s^{-1}, F_A is proportional to square of speed.

Note:

No.	Solution
	Choose at two points whose coordinates are well-spaced out along the graph line. Where a constant of proportionality is to be determined for each point, advise to use 3 significant figure answers for drawing conclusions.
2(a)	Let the mass of the satellite be m.